



CS-17 · GIOU FINAL PRESENTATION

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PROJECT 17 · INTERACTIVE COMPUTER GRAPHICS TOOL

GloU Visualisation *Tool*

An interactive 3D application for creating, manipulating and comparing cuboids while computing and visualising GloU in real time.

RYAN TAN · 32504462 / YAOZE GUO · 31469853 / TIM HUANG · 33867755 / HARRY MESSAGE · 33965358

NATE · TIM · RYAN

SLIDES · B STATIC · ESC INDEX

One product story, three demo handoffs.

NATE · INTRO + DEMO 1

Project definition

Project team, topic, main goals, target users, then object creation and manipulation.

TIM • USER NEEDS + DEMO 2

Use-case proof

Use cases, background research, use-case testing approach, 2D/3D modes, Analyse workflow and metric explanation.

RYAN • DEMO 3 + CONCLUSION

Status and lessons

Save/load, extra features, achieved and remaining scope, issues encountered, teamwork and project management learning.

TEAM	TOPIC	USERS	TESTING	DEMO	CONCLUSION
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What are IoU and GloU?

IOU INTERSECTION OVER UNION

How much do two boxes overlap?

IoU compares the shared area or volume against the total combined area or volume of two objects.

- High IoU means strong overlap.
- Zero IoU means no overlap.
- Useful, but silent when boxes are separate.

GIOU GENERALISED IOU

How good is the spatial match?

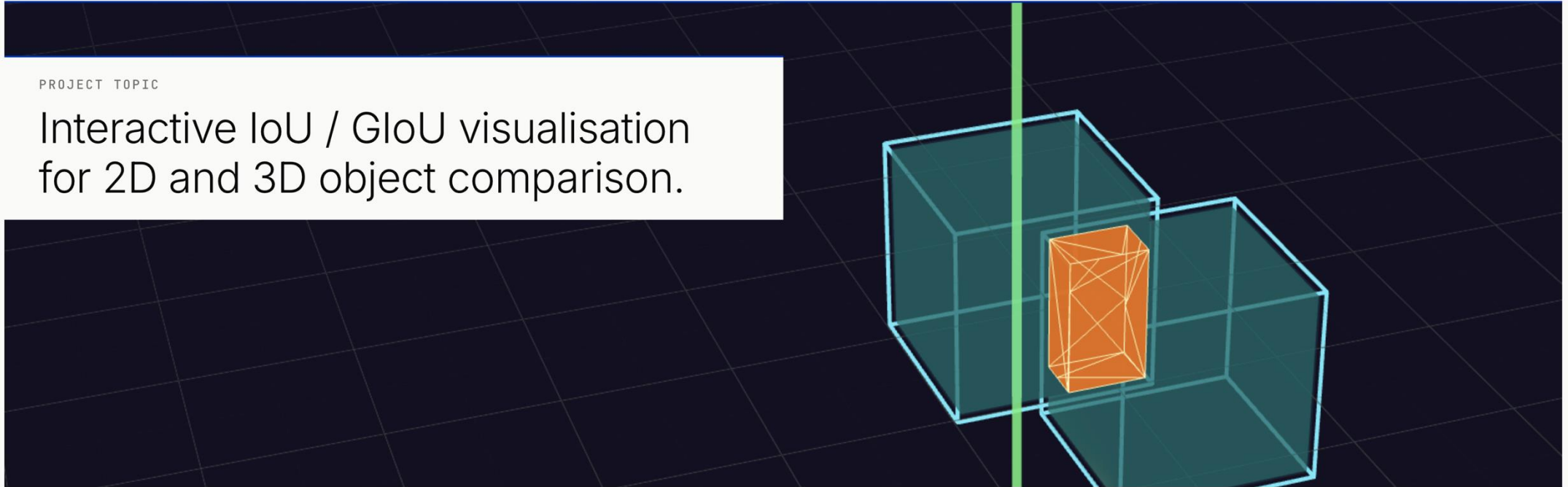
GIoU extends IoU by adding a penalty based on the smallest enclosing box, so it remains meaningful even when objects do not overlap.

- Considers overlap and enclosure.
- Changes even in non-overlap cases.
- Better for teaching spatial metric behaviour.

In this presentation, the key learning loop is: move or resize cuboids → observe IoU/GIoU change → inspect the explanation in Analyse.

PROJECT TOPIC

Interactive IoU / GloU visualisation for 2D and 3D object comparison.



The main goal is to make object-overlap metrics teachable, visible and repeatable through direct manipulation, live calculation, explanation and scene management.

01

Teach

Help learners understand metric behaviour spatially.

02

Interact

Create, move, rotate and scale objects.

03

Compute

Calculate IoU/GIoU as the scene changes.

04

Explain

Connect values to formulas and selected objects.

05

Repeat

Save and reload scenes for reusable examples.

Add, randomise, select, move, rotate and scale.

2D/3D views, selected pair, formula and live metric.

Save/load locally and show extra workflow support.

Achieved, remaining, issues and lessons.

Latest RTM IDs are used here.

Create and transform cuboids.

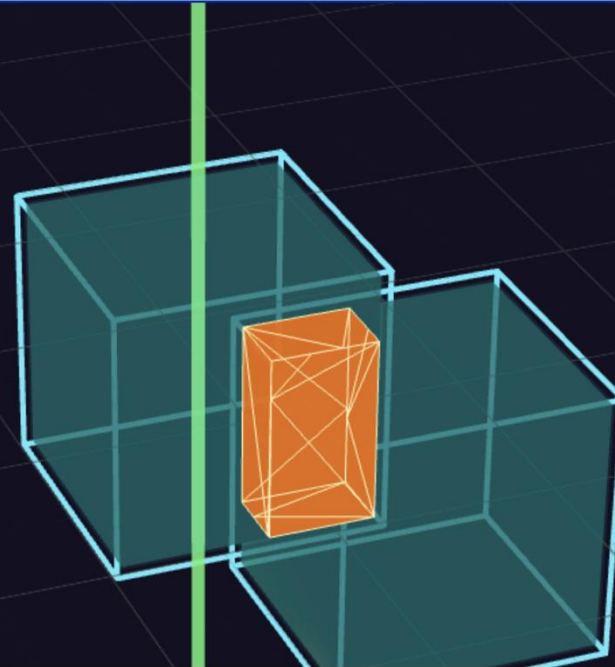
Calculate IoU/GIoU.

Display and explain results.

Auto-update values live.

Save and load scenes.

Create, select and transform cuboids.



CREATE

R2

TRANSFORM

R3

RANDOM

R7

Generate random cuboids for fast examples.

Educator-led explanation

- Shows overlap and non-overlap visually.
- Connects movement to metric change.
- Supports live explanation instead of static diagrams.

Student-driven experiments

- Encourages trial-and-error learning.
- Supports repeatable scene setups.
- Can extend to practitioner metric exploration.

Rezatofighi et al. define GloU for overlap and non-overlap cases using the smallest enclosing box.

Static equations and diagrams do not give beginners enough spatial intuition.

Users need manipulation, live values and explanation in one interface.

Each requirement maps to tests that verify functional delivery.

Delivered features are tested and issues are logged before the next cycle.

Target users perform key tasks such as creating cuboids and observing metric changes.

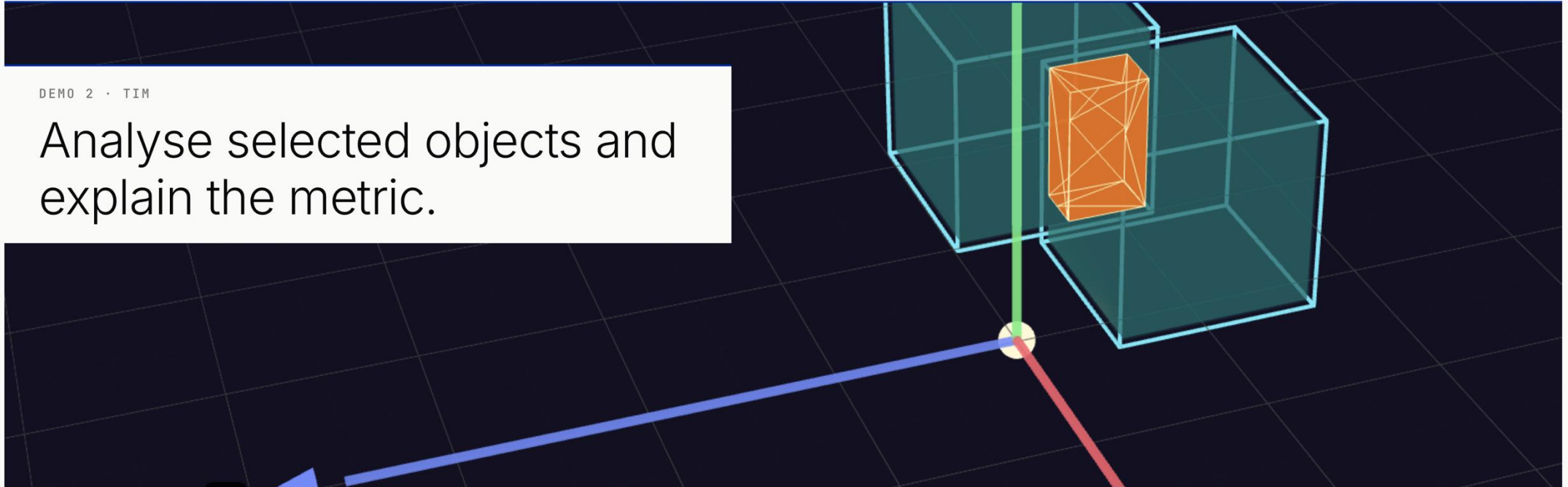
Pre-computed IoU/GIoU examples are compared with tool output.

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The testing approach checks usability, feature delivery and calculation correctness: create object → manipulate object → observe metric change → inspect explanation.

DEMO 2 · TIM

Analyse selected objects and explain the metric.



Switch between 2D and 3D views, then open Analyse to show pair selection, formula explanation, numeric result and live update.

MODE

R11

Switch between 2D and 3D viewing modes.

ANALYSE

R1

Calculate IoU/GIoU for compared objects.

EXPLAIN

R4-R6

Display results and auto-update values live.

Scene persistence

Safe recovery

Workflow support

Show import support, clearer controls, status messages and usability improvements where relevant.

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FINAL STATUS

RYAN: ACHIEVED / REMAINING / ISSUES

Core goals achieved.

R8/N7 remain.

The main cuboid, metric, UI, 2D/3D, import and persistence workflows are delivered; only the authentication-related R8 and N7 items are not fully closed.

HONEST SCOPE CLOSURE

STATUS SUMMARY

ACHIEVED / REMAINING / OPEN ISSUES

ACHIEVED / SIGNED-OFF

R1-R7, R9 and R11-R17 are delivered across metric calculation, object interaction, 2D/3D views, visual feedback and save/load.

RECENTLY COMPLETED

R10 and N1-N6 are treated as recently completed polish and support work: import, performance, usability and data reliability.

REMAINING SCOPE / ISSUES

R8 and N7 are authentication-related and not fully closed; open issues are mostly performance, accessibility polish and maintainability cleanup.

CONCLUSION MESSAGE: CORE DELIVERY ACHIEVED, REMAINING WORK BOUNDED.

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SPECIFIC EVIDENCE / ANECDOTE

Clear ownership mattered most when the team changed.

2

ACTIVE MEMBERS LAST YEAR

The project previously dropped to only two active members, forcing stronger shared awareness.

1

COMMUNICATION LOST THIS YEAR

The remaining team had to stay informed and support each other to avoid delays.

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ISSUES FIXED THIS REVIEW CYCLE

Review findings were converted into implementation items across UI, accessibility, state and interactions.

KEY MESSAGE

Effective teamwork depended on understanding each other's responsibilities and being prepared to support one another when needed.

EXAMPLES FIXED

MetricsTab display behaviour, Analyse mode selection logic, Add Cuboid validation, keyboard shortcuts, accessibility roles and labels, responsive bars, clearer comparison status messages.

RYAN · PROJECT MANAGEMENT LESSON

RTM + GANTT + RISK REGISTER + REVIEW NOTES

Concrete artefacts made Agile usable.

KEY MESSAGE

Effective project management meant using the RTM, Gantt chart, risk register and review notes as one control system.

ISSUES ENCOUNTERED

Remaining work is bounded: R8/N7 plus performance, accessibility polish and maintainability cleanup.

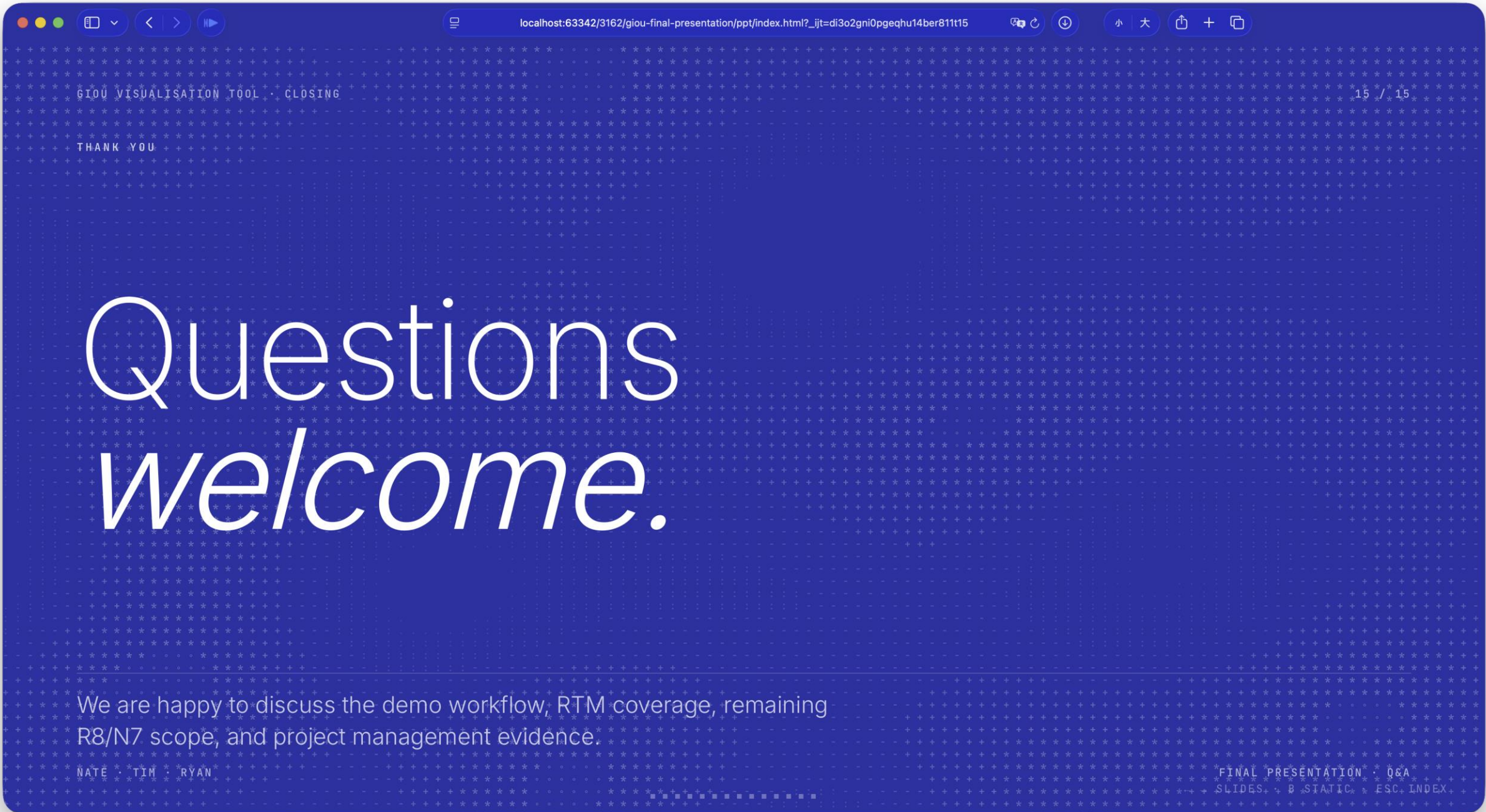
SIMULATED GANTT VIEW

Built from current RTM status and known sprint progress.

W1-W12

WORKSTREAM	1	2	3	4	5	6	7	8	9	10	11	12
RTM + SCOPE	DEFINE / SIGN-OFF				TRACK CHANGES							
CORE 3D		R2/R3/R7					REVIEW POLISH					ANALYSE
		R1/R4/R5/R6							PERFORMANCE		2D / IMPORT	
			R10-R15						NFR		SAVE / LOAD	
			R9/N6					VALIDATION		FINAL DEMO		
						REHEARSE / PRESENT						
RTM Scope control	GANTT Schedule visibility				RISK Owners + mitigation				NOTES Review closure			

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Questions *welcome.*

We are happy to discuss the demo workflow, RTM coverage, remaining R8/N7 scope, and project management evidence.